

Requirement Analysis Report

1. Business Scenario

A college library manages thousands of books and hundreds of students every semester. The current manual system makes it difficult to track books, issue and return records, and student information.

The proposed **Library Management Database System** will maintain all library data in one centralized database. The system will help librarians manage books, students, borrowing, returns, and fines efficiently while reducing paperwork and improving accuracy.

2. Core Entities and Their Purpose

Member

Stores student or staff information who can borrow books.

Attributes

- Member ID
 - Name
 - Email
 - Mobile Number
 - Department
 - Registration Date
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Book

Stores information about books available in the library.

Attributes

- Book ID
 - Book Title
 - Author
 - ISBN
 - Category ID
 - Available Copies
-

Category

Groups books into different categories.

Attributes

- Category ID
 - Category Name
 - Description
-

Librarian

Stores details of library staff.

Attributes

- Librarian ID
 - Name
 - Email
 - Contact Number
-

Issue

Stores information when a member borrows a book.

Attributes

- Issue ID
 - Member ID
 - Book ID
 - Issue Date
 - Due Date
-

Return

Stores return details.

Attributes

- Return ID
 - Issue ID
 - Return Date
 - Fine Amount
-

Fine

Stores fine payment information.

Attributes

- Fine ID
 - Return ID
 - Fine Amount
 - Payment Status
-

Reservation

Stores book reservation details.

Attributes

- Reservation ID
 - Member ID
 - Book ID
 - Reservation Date
-

Review

Stores member reviews about books.

Attributes

- Review ID
 - Member ID
 - Book ID
 - Rating
 - Comments
-

3. Major Business Processes

- Member Registration
- Book Management
- Category Management
- Book Issue
- Book Return
- Fine Calculation

- Reservation Management
 - Book Review Management
 - Report Generation
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4. Stakeholders and Their Roles

Stakeholder	Role
Librarian	Manages books, issues, returns, and members
Student	Searches books, borrows books, submits reviews
Library Administrator	Maintains the complete database and generates reports
College Management	Monitors library usage and reports

5. Functional Requirements

- User Registration
 - Login Authentication
 - Add/Edit/Delete Books
 - Search Books
 - Manage Categories
 - Borrow Books
 - Return Books
 - Calculate Fine
 - Reserve Books
 - Add Reviews
 - Generate Reports
 - Manage Members
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6. Non-Functional Requirements

- User-friendly Interface
 - High Security
 - Fast Performance
 - Data Accuracy
 - Database Backup
 - Reliability
 - Scalability
 - Availability
 - Easy Maintenance
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7. Project Scope

The system includes:

- Member Management
 - Book Management
 - Category Management
 - Issue and Return Management
 - Reservation Management
 - Fine Management
 - Book Review System
 - Report Generation
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8. Assumptions

- Every member has a unique Member ID.
 - Every book has a unique Book ID.
 - Members must register before borrowing books.
 - Books are available only if stock exists.
 - Fine is calculated automatically for late returns.
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9. Constraints

- Internet connection is required.
- Only authorized librarians can modify records.
- Members cannot borrow unavailable books.
- Database should maintain accurate records.
- System follows college library policies.